Online Appendix

OA Applications

OA.1 Network Expansion: Recruiting New Faculty

Illustration: $n = 2$ plus one entrant As an example, consider a network consisting of two existing members (agent 1 and agent 2) and one new agent (agent 3). For simplicity, we assume that the variance of the link is one for all agents; that is, all the diagonal entries of each $\overline{\mathbf{B}}_i$ are equal to one for $i = 1, 2, 3$. Consequently, each entry of $\overline{\mathbf{B}}_i$ matrix represents a correlation coefficient of the influences toward agent i. We assume that agent 1's influence on their own outcome is uncorrelated with agent 2's influence on agent 1's outcome (i.e., $(\mathbf{B}_1)_{12} = 0$). In contrast, we assume that agent 1's influence on agent 2's outcome is correlated with agent 2's influence on agent 2's outcome (i.e., $(\mathbf{B}_2)_{12} = \rho \in [-1, 1]$), and the exact correlation coefficient ρ is known to the DM. Finally, we suppose that all the other entries are unknown to the DM. Under these conditions, the following matrices illustrate the network uncertainty:

$$\overline{\mathbf{B}}_1 = \begin{array}{c} \mathbf{B}_1 \\ 1 \\ 2 \\ 3 \end{array} \begin{array}{c} \begin{array}{ccc} 1 & 2 & 3 \end{array} \\ \begin{bmatrix} 1 & 0 & ? \\ 0 & 1 & ? \\ ? & ? & 1 \end{bmatrix} \end{array}, \quad \overline{\mathbf{B}}_2 = \begin{array}{c} \mathbf{B}_2 \\ 1 \\ 2 \\ 3 \end{array} \begin{array}{c} \begin{array}{ccc} 1 & 2 & 3 \end{array} \\ \begin{bmatrix} 1 & \rho & ? \\ \rho & 1 & ? \\ ? & ? & 1 \end{bmatrix} \end{array}, \quad \text{and} \quad \overline{\mathbf{B}}_3 = \begin{array}{c} \\ 1 \\ 2 \\ 3 \end{array} \begin{array}{c} \begin{array}{ccc} 1 & 2 & 3 \end{array} \\ \begin{bmatrix} 1 & ? & ? \\ ? & 1 & ? \\ ? & ? & 1 \end{bmatrix} \end{array}.$$

There are two notable features in the examples of $\overline{\mathbf{B}}_3$ and the other two matrices. First, for any agent $i = 1, 2, 3$ in the network, all entries representing the correlation coefficient between the influence of the new agent 3 on i and the influence of another existing agent $j = 1, 2$ on i are undetermined; that is, $(\overline{\mathbf{B}}_i)_{j3}$ is undetermined. In the illustrated matrices, this feature is reflected by all the entries in the third row or column being undetermined and represented by the question mark (i.e., '?'). This highlights the first source of uncertainty introduced by the network expansion: the DM lacks information on how the new agent's influence on a given agent correlates with the influence exerted by other existing agents on that agent.

Second, the correlation coefficients of the existing members' influences on agents are asymmetric. Specifically, the influences of existing members on existing agents 1 and 2 are fully known. This is illustrated by the fact that, for agent $i = 1, 2$, the principal submatrix $\mathbf{B}_i$ of $\overline{\mathbf{B}}_i$, highlighted by a blue solid-line box in the illustrated matrices, has all its entries determined. In contrast, the influences of the existing agents on the new agent 3 are unknown. This highlights the second source of uncertainty introduced by the network expansion: the DM lacks information on how the existing agents' influences on the new agent are correlated with one another.

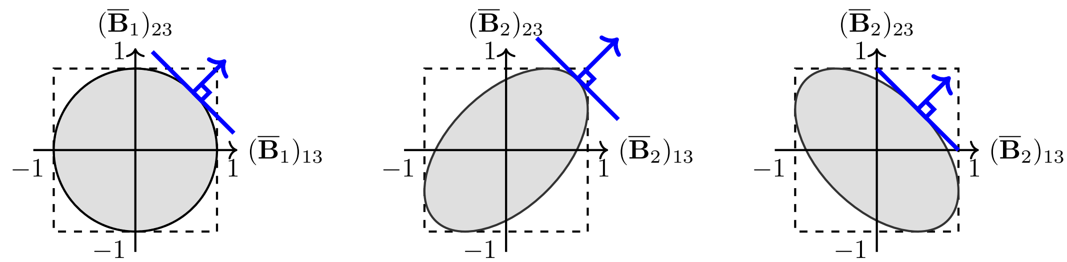
Consequently, all the off-diagonal entries in $\overline{\mathbf{B}}_3$ are undetermined and represented by the question mark in the illustrated matrices. This implies that, in the worst-case scenario, Nature's choice of $\overline{\mathbf{B}}_3$ presents the rank-1 structure, as established in Theorem 1. Therefore, the remaining analysis focuses on characterizing undetermined entries in $\overline{\mathbf{B}}_i$ for $i = 1, 2$ within an appropriate uncertainty set.

By linearity, we have $\langle \overline{\mathbf{x}}, \overline{\mathbf{B}}\overline{\mathbf{x}} \rangle = \langle \overline{\mathbf{x}}, \overline{\mathbf{B}}_1\overline{\mathbf{x}} \rangle + \langle \overline{\mathbf{x}}, \overline{\mathbf{B}}_2\overline{\mathbf{x}} \rangle + \langle \overline{\mathbf{x}}, \overline{\mathbf{B}}_3\overline{\mathbf{x}} \rangle$. Since none of the entries of $\overline{\mathbf{x}}$ is zero, Theorem 1 shows the unique $\overline{\mathbf{B}}_3^*$ is given by $\overline{\mathbf{B}}_3^* = \sigma_3^2(\mathbf{w}_3 \otimes \mathbf{w}_3)$ with $\sigma_3^2 = \sum_{j=1}^{3} \mathbf{v}_{3j}^2 = 3$, $\mathbf{w}_3 = \frac{\mathbf{q}_3}{\|\mathbf{q}_3\|}$, and $\mathbf{q}_3 = (s(\overline{x}_1), s(\overline{x}_2), s(\overline{x}_3))^\mathsf{T}$. As a result, $\overline{\mathbf{B}}_3^*$ is a rank-1 matrix.

On the other hand, the uniqueness of the worst-case scenario for the existing agents $i = 1, 2$ stems from the strict convexity of the uncertainty set, which contrasts with the reasoning behind the uniqueness of $\overline{\mathbf{B}}_3$. To determine $\overline{\mathbf{B}}_1^*$, observe that $\langle \overline{\mathbf{x}}, \overline{\mathbf{B}}_1\overline{\mathbf{x}} \rangle = 2\overline{x}_3\left((\overline{\mathbf{B}}_1)_{13}\overline{x}_1 + (\overline{\mathbf{B}}_1)_{23}\overline{x}_2\right) + \text{constant}$, where $(\overline{\mathbf{B}}_1)_{13} = \mathrm{Corr}(\mathbf{G}_{11}, \mathbf{G}_{13}) \in [-1, 1]$ and $(\overline{\mathbf{B}}_1)_{23} = \mathrm{Corr}(\mathbf{G}_{12}, \mathbf{G}_{13}) \in [-1, 1]$. The dashed square in Figure OA1-(a) represents these restrictions of the possible values of $(\overline{\mathbf{B}}_1)_{13}$ and $(\overline{\mathbf{B}}_1)_{23}$. Additionally, Nature's choice of uncertainty is constrained by the requirement that $\overline{\mathbf{B}}_1$ must be positive semi-definite, which holds if and only if $\det(\overline{\mathbf{B}}_1) \geq 0$ because $\mathbf{B}_1 \in \mathrm{PD}^2$ and $(\overline{\mathbf{B}}_1)_{33} > 0$. The Schur Complement theorem (Meyer, 2010) states that $\det(\overline{\mathbf{B}}_1) \geq 0$ if and only if $(\overline{\mathbf{B}}_1)_{33} - ((\overline{\mathbf{B}}_1)_{13}, (\overline{\mathbf{B}}_1)_{23})\mathbf{B}_1^{-1}((\overline{\mathbf{B}}_1)_{13}, (\overline{\mathbf{B}}_1)_{23})^\mathsf{T} \geq 0$, which is equivalent to $(\overline{\mathbf{B}}_1)_{13}^2 + (\overline{\mathbf{B}}_1)_{23}^2 \leq 1$, represented by the gray unit disk in Figure OA1-(a) contained in the dashed square. Consequently, the uncertainty set is strictly convex and compact.[17] The worst-case scenario for agent 1 must therefore be chosen within this uncertainty set.

Since the uncertainty set is strictly convex, the gradient of $\langle \overline{\mathbf{x}}, \overline{\mathbf{B}}_1\overline{\mathbf{x}} \rangle$ with respect to $\left((\overline{\mathbf{B}}_1)_{13}, (\overline{\mathbf{B}}_1)_{23}\right)$ must be orthogonal to the boundary of the uncertainty set. The assumption of non-zero entries in $\overline{\mathbf{x}}$ implies that the gradient (represented by the

[17] Our proof in Appendix C addresses a more general condition, and the strict convexity of the uncertainty set is not a direct consequence of the Schur Complement theorem.



(a) uncertainty set for agent 1 (b) uncertainty set for agent 2 if $\rho = -0.5$ (c) uncertainty set for agent 2 if $\rho = 0.5$

Figure OA1: Illustration of the worst-case scenario. In each figure, the gradient of Nature's objective function at the unique worst-case scenario is orthogonal to the uncertainty set denoted by the gray elliptical disk.

blue arrow in Figure OA1) is not vanishing. Consequently, there is a unique optimal choice $\overline{\mathbf{B}}_1^*$, in which $\nabla\langle \overline{\mathbf{x}}, \overline{\mathbf{B}}_1 \overline{\mathbf{x}}\rangle\big|_{\overline{\mathbf{B}}_1 = \overline{\mathbf{B}}_1^*} = 2\overline{x}_3(\overline{x}_1, \overline{x}_2)$ is orthogonal to the boundary of the uncertainty set as illustrated in Figure OA1-(a). At the optimal choice, a trade-off arises among the correlation coefficients due to the strict convexity of the uncertainty set, which results from the partial correlation information constraint $\mathbf{B}_1$.

The worst-case scenario for agent 2, $\overline{\mathbf{B}}_2^*$, is also uniquely determined by the strict convexity of the uncertainty set. However, it is worth examining how the shape of uncertainty set changes due to the presence of the correlation among the influences toward agent 2. The dashed square in Figure OA1-(b) represents the constraints that $(\overline{\mathbf{B}}_2)_{13} = \mathrm{Corr}(\mathbf{G}_{21}, \mathbf{G}_{23}) \in [-1, 1]$ and $(\overline{\mathbf{B}}_2)_{23} = \mathrm{Corr}(\mathbf{G}_{22}, \mathbf{G}_{23}) \in [-1, 1]$. Again, Nature's choice of uncertainty faces another restriction: $\overline{\mathbf{B}}_2$ must be positive semi-definite, which holds if and only if, by the Schur Complement theorem,

$$\begin{bmatrix} (\overline{\mathbf{B}}_2)_{13} & (\overline{\mathbf{B}}_2)_{23} \end{bmatrix} \begin{bmatrix} 1 & -\rho \\ -\rho & 1 \end{bmatrix} \begin{bmatrix} (\overline{\mathbf{B}}_2)_{13} \\ (\overline{\mathbf{B}}_2)_{23} \end{bmatrix} \leq 1 - \rho^2.$$

The pairs of $\big((\overline{\mathbf{B}}_2)_{13}, (\overline{\mathbf{B}}_2)_{23}\big)$ satisfying the above inequality forms an elliptical disk as illustrated in Figure OA1-(b),(c). For each $\rho \in [-1, 1]$, there are two invariant principal components (i.e., eigenvectors) $(1, 1)^\intercal$ and $(1, -1)^\intercal$, with corresponding non-negative eigenvalues $1 - \rho$ and $1 + \rho$, respectively. The uncertainty set for $\rho = -0.5$ is shown as the rotated gray elliptical disk in Figure OA1-(b), and the uncertainty set for $\rho = 0.5$ is illustrated in Figure OA1-(c). In both figures, the uncertainty sets are strictly convex.

In summary, the strict convexity of the uncertainty sets yields a unique worst-case scenario $\overline{\mathbf{B}}_i^*$ for each $i = 1, 2, 3$, and moreover, $\overline{\mathbf{B}}_2^*$ can vary depending on the correlation coefficient ρ. If all the entries of $\overline{\mathbf{x}}$ are positive, the gradient of Nature's objective function at the worst-case scenario is orthogonal to the uncertainty set and lies in the first quadrant. As a result, the uncertainty set yields lower values of $(\overline{\mathbf{B}}_2^*)_{13}$ and $(\overline{\mathbf{B}}_2^*)_{23}$ as the correlation ρ increases, as shown in Figure OA1-(c). On the other hand, if $\overline{\mathbf{x}}$ includes a negative entry (e.g., $\overline{\mathbf{x}}_1 > 0$ and $\overline{\mathbf{x}}_2 < 0$), the worst-case scenario values for $(\overline{\mathbf{B}}_2)_{13}$ and $(\overline{\mathbf{B}}_2)_{23}$ may increase in magnitude as ρ increases, as in Figure OA1-(b).

OB Higher-Order Interactions

So far, we have addressed the robust intervention problem in the context of one-shot interactions among agents. However, in many studies in network economics, the focus shifts to settings where agents interact repeatedly or infinitely, leading to long-run equilibrium outcomes. Such scenarios are commonly analyzed using the equilibrium representation $(\mathbf{I} - \delta\mathbf{G})^{-1}$, where $\delta > 0$ captures the strength of the network effect.

If the spectral radius of $\delta\mathbf{G}$ is less than one, ensured when δ is sufficiently small, the equilibrium can be expressed as a power series: $(\mathbf{I} - \delta\mathbf{G})^{-1} = \sum_{k=0}^{\infty}(\delta\mathbf{G})^k$. In this representation, $\mathbf{G}_{ij}^k$ represents the weighted sum of all walks of length k from agent j to agent i, capturing the k-th order influence of agent j's allocation on agent i's outcome.[18] This power series representation aggregates the influence dynamics across all possible walks within the network.

The critical question, then, is how the DM can incorporate these higher-order influences and the associated higher-order uncertainties into the design of a robust intervention. By accounting for these dynamics, the DM can address the challenges of designing effective strategies in networks where agents' interactions propagate over time and across multiple pathways. Motivated by this, we now consider an extension of our model to incorporate higher-order uncertainty under additional assumptions.

First, we assume that with a second-order approximation of $(\mathbf{I} - \delta\mathbf{G})^{-1} \approx (\mathbf{I} + \delta\mathbf{G} + \delta^2\mathbf{G}^2)$, the DM minimizes the following objective:

$$\mathrm{E}\left[\|(\mathbf{I} + \delta\mathbf{G} + \delta^2\mathbf{G}^2)\mathbf{x} - \mathbf{z}\|^2\right]. \tag{OB.1}$$

Second, we assume $\mathbf{G}_{ii} = 0$ for all $i \in N$. This assumption is often valid in contexts such as network games or supply chain network models, where the zero-

[18] A walk of length k from node j to i is a sequence of nodes $(i_0, i_1, i_2, \ldots, i_k)$, such that $i_0 = j$, $i_k = i$, and nodes in the sequence need not be distinct (Jackson, 2010).

order interaction term $\mathbf{I}$ captures an agent's self-influence after normalization. Third, we assume statistical independence between the rows of $\mathbf{G}$, meaning that the influence on an agent is independent of the influence on any other agents. For instance, in a social learning model, agent i's weight on agent k is independent of agent j's weight on k.

Under these assumptions, the adversarial Nature's choice of the worst-case scenario is characterized by a unique rank-1 covariance matrix $\mathbf{B}_i^*$ for each agent i. To simplify exposition, we assume that $\mathbf{m}_{ij} = \mathrm{E}[\mathbf{G}_{ij}] = 0$ for all $i, j \in N$, allowing us to focus solely on the impact of uncertainty.[19] Under this assumption, the expected squared distance between agent i's final outcome and the target outcome z_i is

$$\mathrm{E}\left[|(\mathbf{I} + \delta\mathbf{G} + \delta^2\mathbf{G}^2)_i\mathbf{x} - z_i|^2\right] = x_i^2 + \delta^2\langle\mathbf{x}, \mathbf{B}_i\mathbf{x}\rangle + \sum_{k=1}^{n} \delta^4\mathbf{v}_{ik}^2\langle\mathbf{x}, \mathbf{B}_k\mathbf{x}\rangle + z_i^2 - 2z_ix_i. \tag{OB.2}$$

The term $\delta^2\langle\mathbf{x}, \mathbf{B}_i\mathbf{x}\rangle$ in expression OB.2 arises from $(\delta\mathbf{G})_i\mathbf{x}$, capturing the uncertainty generated by the first-order interactions among the agents with respect to agent i.

Another term, $\delta^4\sum_{k=1}^{n}\mathbf{v}_{ik}^2\langle\mathbf{x}, \mathbf{B}_k\mathbf{x}\rangle$, accounts for the second-order interactions affecting agent i's final outcome. For these second-order interactions, consider pairs of agents s and k with intermediate agents s' and k' directly influencing agent i. The second-order interactions from s to i and from k to i are represented by $\mathbf{G}_{is'}\mathbf{G}_{s's}$ and $\mathbf{G}_{ik'}\mathbf{G}_{k'k}$, respectively. If $s' \neq k'$, the correlation between these second-order interactions is zero due to the independence in link formation among the agents' interactions, captured by the computation $\mathrm{E}\left[\mathbf{G}_{is'}\mathbf{G}_{s's}\mathbf{G}_{ik'}\mathbf{G}_{k'k}\right] = \mathrm{E}\left[\mathbf{G}_{is'}\mathbf{G}_{ik'}\right]\mathrm{E}\left[\mathbf{G}_{s's}\right]\mathrm{E}\left[\mathbf{G}_{k'k}\right] = 0$.

Consequently, the second order effect arises only when there is a common intermediate agent (i.e., $k' = s'$). For each intermediate agent, the correlations among walks of length 2 that share agent k' as the common intermediate agent are amplified by the discounted variance $\delta^4\mathbf{v}_{ik'}^2$. For example, in Figure OA2, the solid blue arrows directed toward intermediate agent 2 influence agent 1's outcome and are independent of the other arrows directed toward a distinct agent, such as the dashed green arrows leading to intermediate agent 3. Then, the impact of the blue arrows on agent 1 is weighted by the discounted variance term $\delta^4\mathbf{v}_{12}^2$, while the impact of the green arrows on agent 1 is weighted by the discounted variance term $\delta^4\mathbf{v}_{13}^2$.

[19] The proof of Proposition A4 does not rely on this zero mean influence of the agents.

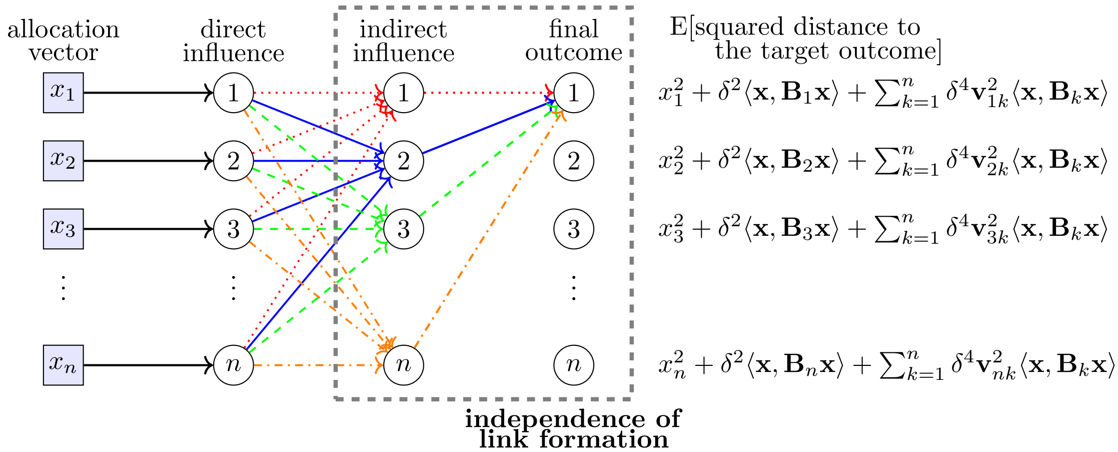


Figure OA2: Illustration of second-order uncertainty generation in network. For simplicity, here we assume that $\mathbf{z} = \mathbf{0}$.

By summing the squared distances for all agents, we obtain

$$\mathrm{E}\left[\|(\mathbf{I} + \delta \mathbf{G} + \delta^2 \mathbf{G}^2)\mathbf{x} - \mathbf{z}\|^2\right] = \langle \mathbf{x}, \mathbf{x} \rangle + \sum_{i=1}^{n} w_i \langle \mathbf{x}, \mathbf{B}_i \mathbf{x} \rangle - 2\langle \mathbf{z}, \mathbf{x} \rangle + \langle \mathbf{z}, \mathbf{z} \rangle, \qquad \text{OB.3}$$

where $w_i = \delta^2(1 + \delta^2 \sum_{k=1}^n \mathbf{v}_{ki}^2) > 0$. The above expression remains linear in each $\mathbf{B}_i$, similar to expression (3) in the main model without higher-order considerations. As detailed in Theorem 1, this implies that the adversarial Nature's selection of the worst-case scenario satisfies the rank-1 property for all agents. The following proposition summarizes this result.

Proposition A4 *With the higher-order consideration with the objective function OB.2, there exists a unique worst-case scenario $\mathbf{B}_i^*$ for each agent $i \in N$, and it is rank-1.*

The uniqueness of the worst-case scenario, in turn, allows the DM to solve her robust optimal intervention problem by selecting her intervention based on the first-order condition under the worst-case scenario, in a manner analogous to Theorem 1.

We conclude this subsection by noting that for k-th order interaction considerations with $k \geq 3$, the linearity in the objective with respect to the uncertainties $(\mathbf{B}_i)_{i \in N}$ no longer holds, even under the assumption of independent link formation. The reason is that, even with independent link formation, higher-order interactions among influences on an agent can emerge. For example, consider walks of length 3, such as $(3, 2, 1, 2)$ and $(4, 2, 1, 2)$, which differ only in the initial agent. In this case, the interaction among the influences $\mathbf{G}_{23}$, $\mathbf{G}_{24}$ and $\mathbf{G}_{21}$ in the covariance term may introduce nonlinear effects. As a result, the rank-1 property and the uniqueness of the worst-case scenario may no longer hold. We leave the analysis of higher-order interactions and robust intervention design for future research.

Proof of Proposition A4

Proof. For simplicity, we prove the proposition by assuming that the mean influence of the link $\mathbf{G}_{ij}$ is zero for all $i, j \in N$. Then, we provide a general proof without the assumption.

For each $i \in N$, we find that

$$
\begin{aligned}
|(\mathbf{I}+\delta\mathbf{G}+\delta^2\mathbf{G}^2)_i\mathbf{x}-z_i|^2 &= \left|x_i+\delta\sum_{j=1}^{n}\mathbf{G}_{ij}x_j+\delta^2\sum_{1\le k,l\le n}\mathbf{G}_{ik}\mathbf{G}_{kl}x_l-z_i\right|^2 \\
&= x_i^2+\left(\delta\sum_{j=1}^{n}\mathbf{G}_{ij}x_j\right)^2+\left(\delta^2\sum_{1\le k,l\le n}\mathbf{G}_{ik}\mathbf{G}_{kl}x_l\right)^2+z_i^2+2x_i\left(\delta\sum_{j=1}^{n}\mathbf{G}_{ij}x_j\right) \\
&\quad +2x_i\left(\delta^2\sum_{1\le k,l\le n}\mathbf{G}_{ik}\mathbf{G}_{kl}x_l\right)-2x_iz_i+2\left(\delta\sum_{j=1}^{n}\mathbf{G}_{ij}x_j\right)\left(\delta^2\sum_{1\le k,l\le n}\mathbf{G}_{ik}\mathbf{G}_{kl}x_l\right) \\
&\quad -2z_i\left(\delta\sum_{j=1}^{n}\mathbf{G}_{ij}x_j\right)-2z_i\left(\delta^2\sum_{1\le k,l\le n}\mathbf{G}_{ik}\mathbf{G}_{kl}x_l\right). \qquad \text{OB.4}
\end{aligned}
$$

We investigate the expectation of each term in expression OB.4. First, we find that

$$
\mathrm{E}\left[\left(\sum_{1\le k,l\le n}\mathbf{G}_{ik}\mathbf{G}_{kl}x_l\right)\right]=0,\quad \mathrm{E}\left[\left(\sum_{j=1}^{n}\mathbf{G}_{ij}x_j\right)\left(\sum_{1\le k,l\le n}\mathbf{G}_{ik}\mathbf{G}_{kl}x_l\right)\right]=0,
$$
$$
\mathrm{E}\left[\left(\sum_{j=1}^{n}\mathbf{G}_{ij}x_j\right)\right]=0,\quad \mathrm{E}\left[\left(\sum_{1\le k,l\le n}\mathbf{G}_{ik}\mathbf{G}_{kl}x_l\right)\right]=0.
$$

Second, $\mathrm{E}\left[\left(\sum_{j=1}^{n}\mathbf{G}_{ij}x_j\right)^2\right]=\langle\mathbf{x},\mathbf{B}_i\mathbf{x}\rangle$ as in the main model. Third, we observe that

$$
\mathrm{E}\left[\left(\sum_{1\le k,l\le n}\mathbf{G}_{ik}\mathbf{G}_{kl}x_l\right)^2\right]=\mathrm{E}\left[(\sum_{1\le k,l,s,t\le n}\mathbf{G}_{ik}\mathbf{G}_{kl}\mathbf{G}_{is}\mathbf{G}_{st}x_lx_t)\right]
$$
$$
=\sum_{k\ne i,s\ne i,l,t}\mathrm{E}\left[\mathbf{G}_{ik}\mathbf{G}_{is}\right]\mathrm{E}\left[\mathbf{G}_{kl}\mathbf{G}_{st}\right]x_lx_t=\sum_{k\ne i,l,t}\mathrm{E}\left[\mathbf{G}_{ik}^2\right]\mathrm{E}\left[\mathbf{G}_{kl}\mathbf{G}_{kt}\right]x_lx_t=\sum_{k=1}^{n}\mathbf{v}_{ik}^2\langle\mathbf{x},\mathbf{B}_k\mathbf{x}\rangle.
$$

Thus, we obtain expression OB.2 in the main text:

$$\mathrm{E}\left[|(\mathbf{I}+\delta\mathbf{G}+\delta^2\mathbf{G}^2)_i\mathbf{x}-z_i|^2\right] = x_i^2 + \mathrm{E}\left[\left(\delta\sum_{j=1}^{n}\mathbf{G}_{ij}x_j\right)^2 + \left(\delta^2\sum_{1\le k,l\le n}\mathbf{G}_{ik}\mathbf{G}_{kl}x_l\right)^2\right] + z_i^2$$

$$= x_i^2 + \delta^2\langle\mathbf{x},\mathbf{B}_i\mathbf{x}\rangle + \delta^4\sum_{k=1}^{n}\mathbf{v}_{ik}^2\langle\mathbf{x},\mathbf{B}_k\mathbf{x}\rangle + z_i^2 - 2z_ix_i. \tag{OB.5}$$

Finally, by summing the squared distances for all agents, we obtain

$$\mathrm{E}\left[\|(\mathbf{I}+\mathbf{G}+\mathbf{G}^2)\mathbf{x}-\mathbf{z}\|^2\right] = \sum_{i=1}^{n}\left(x_i^2 + \delta^2\langle\mathbf{x},\mathbf{B}_i\mathbf{x}\rangle + \delta^4\sum_{k=1}^{n}\mathbf{v}_{ik}^2\langle\mathbf{x},\mathbf{B}_k\mathbf{x}\rangle - 2z_ix_i + z_i^2\right)$$

$$= \langle\mathbf{x},\mathbf{x}\rangle + \sum_{i=1}^{n}w_i\langle\mathbf{x},\mathbf{B}_i\mathbf{x}\rangle - 2\langle\mathbf{z},\mathbf{x}\rangle + \langle\mathbf{z},\mathbf{z}\rangle, \tag{OB.6}$$

where $w_i = \delta^2 + \delta^4\sum_{k=1}^{n}\mathbf{v}_{ki}^2 > 0$ as in expression OB.3 in the main text.

Since the DM's objective function OB.6 is linear in each $\mathbf{B}_i$, the rank-1 property and the uniqueness of $\mathbf{B}_i^*$ are held by Theorem 1. Therefore, the worst-case scenario $\mathbf{B}^* = \sum_{i=1}^{n}\mathbf{B}_i^*$ is unique.

We now prove the proposition without the assumption. To economize notation, without loss of generality, we let $\delta = 1$ because it does not affect the linearity of the DM's objective function. We let $\overline{\mathbf{G}} = \mathrm{E}[\mathbf{G}]$ and write $\mathbf{G} = \overline{\mathbf{G}} + \mathbf{U}$. We denote by $\mathbf{m}_i$ and $\mathbf{m}^i$ the ith row vector and column vector of $\overline{\mathbf{G}}$, respectively. Recall that $\mathbf{M}_i = \mathbf{m}_i\otimes\mathbf{m}_i$, and $\mathrm{E}[\mathbf{U}_i\mathbf{U}_i^\intercal] = \mathbf{B}_i$. Since we assume that $\mathbf{G}_{ii} = 0$, we have $\mathbf{U}_{ii} = \mathbf{m}_{ii} = 0$. We calculate that for each $i \in N$,

$$\mathrm{E}\left[\left(\sum_{j=1}^{n}\mathbf{G}_{ij}x_j\right)^2\right] = \mathrm{E}\left[\left((\mathbf{m}_i+\mathbf{U}_i)^\intercal\mathbf{x}\right)^2\right] = \langle\mathbf{x},\mathbf{B}_i\mathbf{x}\rangle + \langle\mathbf{x},\mathbf{M}_i\mathbf{x}\rangle.$$

In addition, it follows that for each $i \in N$,

$$\mathrm{E}\left[\left(\sum_{1\le k,l\le n}\mathbf{G}_{ik}\mathbf{G}_{kl}x_l\right)^2\right] = \mathrm{E}\left[\sum_{1\le k,l,s,t\le n}\mathbf{G}_{ik}\mathbf{G}_{kl}\mathbf{G}_{is}\mathbf{G}_{st}x_lx_t\right]$$

$$= \sum_{1\le l,t\le n}\sum_{k\ne i,s\ne i}\mathrm{E}\left[\mathbf{G}_{ik}\mathbf{G}_{is}\right]\mathrm{E}\left[\mathbf{G}_{kl}\mathbf{G}_{st}\right]x_lx_t$$

$$= \sum_{1\le l,t\le n}\sum_{k\ne i,s\ne i}(\mathrm{E}[\mathbf{U}_{ik}\mathbf{U}_{is}] + \mathbf{m}_{ik}\mathbf{m}_{is})\,\mathrm{E}([\mathbf{U}_{kl}\mathbf{U}_{st}] + \mathbf{m}_{kl}\mathbf{m}_{st})x_lx_t.$$

We find the following useful expressions for each $i \in N$:

$$\sum_{1\le l,t\le n}\sum_{k\ne i,s\ne i}\mathrm{E}[\mathbf{U}_{ik}\mathbf{U}_{is}]\,\mathrm{E}\left[\mathbf{U}_{kl}\mathbf{U}_{st}\right]x_l x_t = \sum_{k\ne i}\sum_{l,t}\mathrm{E}[\mathbf{U}_{ik}\mathbf{U}_{ik}]\,\mathrm{E}\left[\mathbf{U}_{kl}\mathbf{U}_{kt}\right]x_l x_t = \sum_{k}\mathbf{v}_{ik}^2\langle\mathbf{x},\mathbf{B}_k\mathbf{x}\rangle,$$

$$\sum_{1\le l,t\le n}\sum_{k\ne i,s\ne i}\mathbf{m}_{ik}\mathbf{m}_{is}\,\mathrm{E}\left[\mathbf{U}_{kl}\mathbf{U}_{st}\right]x_l x_t = \sum_{k}\mathbf{m}_{ik}^2\langle\mathbf{x},\mathbf{B}_k\mathbf{x}\rangle,$$

$$\sum_{1\le l,t\le n}\sum_{k\ne i,s\ne i}\mathbf{m}_{kl}\mathbf{m}_{st}\,\mathrm{E}[\mathbf{U}_{ik}\mathbf{U}_{is}]x_l x_t = \sum_{l,t}\langle x_l\mathbf{m}^l,\mathbf{B}_i(x_t\mathbf{m}^\intercal)\rangle = \langle\overline{\mathbf{G}}\mathbf{x},\mathbf{B}_i\overline{\mathbf{G}}\mathbf{x}\rangle,$$

$$\sum_{1\le l,t,k,s\le n}\mathbf{m}_{ik}\mathbf{m}_{is}\mathbf{m}_{kl}\mathbf{m}_{st}x_l x_t = \sum_{l,t}(\overline{\mathbf{G}}_{il}^2 x_l)(\overline{\mathbf{G}}_{it}^2 x_t) = (\overline{\mathbf{G}}^2\mathbf{x})_i^2.$$

Then, we obtain that for each $i \in N$,

$$\mathrm{E}\left[\left(\sum_{1\le k,l\le n}\mathbf{G}_{ik}\mathbf{G}_{kl}x_l\right)^2\right] = \sum_{k=1}^{n}(\mathbf{v}_{ik}^2+\mathbf{m}_{ik}^2)\langle\mathbf{x},\mathbf{B}_k\mathbf{x}\rangle + \langle\overline{\mathbf{G}}\mathbf{x},\mathbf{B}_i\overline{\mathbf{G}}\mathbf{x}\rangle + (\overline{\mathbf{G}}^2\mathbf{x})_i^2,$$

$$\mathrm{E}\left[\sum_{j=1}^{n}\mathbf{G}_{ij}x_j\right] = \sum_{j=1}^{n}\mathbf{m}_{ij}x_j = \langle\mathbf{m}_i,\mathbf{x}\rangle = (\overline{\mathbf{G}}\mathbf{x})_i,$$

$$\mathrm{E}\left[\sum_{1\le k,l\le n}\mathbf{G}_{ik}\mathbf{G}_{kl}x_l\right] = \sum_{k,l}\left(\mathbf{m}_{ik}\mathbf{m}_{kl} + \mathrm{E}\left[\mathbf{U}_{ik}\mathbf{U}_{kl}\right]\right)x_l = (\overline{\mathbf{G}}^2\mathbf{x})_i + 0 = (\overline{\mathbf{G}}^2\mathbf{x})_i,$$

$$\begin{aligned}\mathrm{E}\left[\sum_{1\le j,k,l\le n}\mathbf{G}_{ij}\mathbf{G}_{ik}\mathbf{G}_{kl}x_j x_l\right] &= \sum_{1\le j,l\le n}\sum_{k\ne i}\mathrm{E}\left[\mathbf{G}_{ij}\mathbf{G}_{ik}\right]\mathrm{E}\left[\mathbf{G}_{kl}\right]x_j x_l \\ &= \sum_{1\le j,l,k\le n}\left(\mathrm{E}\left[\mathbf{U}_{ij}\mathbf{U}_{ik}\right] + \mathbf{m}_{ij}\mathbf{m}_{ik}\right)\mathbf{m}_{kl}x_j x_l \\ &= \sum_{l=1}^{n}\left(\langle x_l\mathbf{m}^l,\mathbf{B}_i\mathbf{x}\rangle + \langle x_l\mathbf{m}^l,\mathbf{M}_i\mathbf{x}\rangle\right) \\ &= \langle\overline{\mathbf{G}}\mathbf{x},\mathbf{B}_i\mathbf{x}\rangle + \langle\overline{\mathbf{G}}\mathbf{x},\mathbf{M}_i\mathbf{x}\rangle.\end{aligned}$$

Thus, we combine the above expressions and obtain that for each $i \in N$,

$$\begin{aligned}&\mathrm{E}\left[|(\mathbf{I}+\mathbf{G}+\mathbf{G}^2)_i\mathbf{x} - z_i|^2\right] \\ &= \langle\mathbf{x},\mathbf{B}_i\mathbf{x}\rangle + \sum_{k=1}^{n}(\mathbf{v}_{ik}^2+\mathbf{m}_{ik}^2)\langle\mathbf{x},\mathbf{B}_k\mathbf{x}\rangle + \langle\overline{\mathbf{G}}\mathbf{x},\mathbf{B}_i\overline{\mathbf{G}}\mathbf{x}\rangle + 2\langle\overline{\mathbf{G}}\mathbf{x},\mathbf{B}_i\mathbf{x}\rangle + \langle\mathbf{x},\mathbf{M}_i\mathbf{x}\rangle + (\overline{\mathbf{G}}^2\mathbf{x})_i^2 \\ &\quad + 2\langle\overline{\mathbf{G}}\mathbf{x},\mathbf{M}_i\mathbf{x}\rangle + 2\langle x_i\mathbf{m}_i,\mathbf{x}\rangle + 2x_i(\overline{\mathbf{G}}^2\mathbf{x})_i + x_i^2 - 2\langle z_i\mathbf{m}_i,\mathbf{x}\rangle - 2z_i(\overline{\mathbf{G}}^2\mathbf{x})_i - 2z_i x_i + z_i^2.\end{aligned}$$

We now let $\alpha_k = \sum_{i=1}^{n}(\mathbf{v}_{ik}^2+\mathbf{m}_{ik}^2)$. By summing over i, it follows that

$$\mathrm{E}\left[\|(\mathbf{I}+\mathbf{G}+\mathbf{G}^2)\mathbf{x}-\mathbf{z}\|^2\right] = \sum_{i=1}^{n}\left(\alpha_i\langle\mathbf{x},\mathbf{B}_i\mathbf{x}\rangle + \langle(\overline{\mathbf{G}}+\mathbf{I})\mathbf{x},\mathbf{B}_i(\overline{\mathbf{G}}+\mathbf{I})\mathbf{x}\rangle\right) + Q(\mathbf{x}), \quad \text{OB.7}$$

where $Q(\mathbf{x})$ is a quadratic function of $\mathbf{x}$ not involving the uncertainty matrix $(\mathbf{B}_i)_i$ for any $i \in N$. As such, we can consider Q as an additional cost part of the DM's objective function. It is straightforward to see that the DM's objective function OB.7 is still a linear function of $(\mathbf{B}_i)_i$ as before. Consequently, we conclude that when $(\mathbf{B}_1^*, \ldots, \mathbf{B}_n^*)$ is Nature's best response with respect to a given $\mathbf{x}$ having no zero entry, its ith component $\mathbf{B}_i^*$ is uniquely determined as a rank-1 matrix if and only if the following nondegeneracy condition holds:

$$\alpha_i x_j x_k + \big((\overline{\mathbf{G}} + \mathbf{I})\mathbf{x}\big)_j \big((\overline{\mathbf{G}} + \mathbf{I})\mathbf{x}\big)_k \neq 0 \ \text{ for all } \ 1 \leq j < k \leq n. \qquad \text{OB.8}$$

Therefore, the proposition is proven. ■

OC Microfoundation of the Uncertainty Set

This appendix collects additional interpretations of the uncertainty set $\mathcal{B}$ introduced in Subsection 2.2. Throughout, $\mathbf{U}_i = \mathbf{G}_i^\top - \mathbf{m}_i \in \mathbb{R}^n$ denotes the column vector of row-i deviations, as defined in Subsection 2.1.

OC.1 Bayesian Interpretation: A Set of Priors Consistent with Trusted Moments

Let Π be the set of distributions over $\mathbf{G}$ such that $\mathrm{E}_\pi[\mathbf{G}_{ij}] = \mathbf{m}_{ij}$ and $\mathrm{Var}_\pi(\mathbf{G}_{ij}) = \mathbf{v}_{ij}^2$ for all $i, j \in N$. For any $\pi \in \Pi$, define $\mathbf{B}_i(\pi) = \mathrm{E}_\pi[\mathbf{U}_i \mathbf{U}_i^\top]$. The mapping $\pi \mapsto (\mathbf{B}_i(\pi))_{i \in N}$ induces a set of feasible row covariance matrices contained in $\prod_{i \in N} \mathcal{B}_i$, and Nature's move can be interpreted as selecting a least favorable dependence structure among those consistent with the DM's trusted marginal information. Moreover, $\mathcal{B}_i$ can be viewed as a least-restrictive envelope: for any $\mathbf{B}_i \succeq 0$ with diagonal $(\mathbf{v}_{i1}^2, \ldots, \mathbf{v}_{in}^2)$, there exists a distribution over $\mathbf{U}_i$ with covariance $\mathbf{B}_i$ (for example, multivariate normal), so $\mathcal{B}_i$ contains all covariance structures compatible with the maintained second-moment information and covariance feasibility.

OC.2 Partial-Identification Interpretation: An Identified Set for Dependence

Suppose an econometrician observes data generated by $\mathbf{G}$ and can consistently estimate $\mathbf{m}_{ij}$ and $\mathbf{v}_{ij}^2$ for each link, but cannot identify $\mathrm{Cov}(\mathbf{G}_{ij}, \mathbf{G}_{ik})$ beyond general inequalities implied by feasibility (e.g., Cauchy–Schwarz and positive semidefiniteness). Then $\mathcal{B}_i$ is an identified set of observationally equivalent within-row covariance structures, and $\mathcal{B}$ is the corresponding identified set for aggregated covariance objects that matter for expected loss. In this interpretation, the DM's max–min problem is a minimax decision rule over the identified set, as in robust treatment-choice problems under ambiguity and partial identification.[20]

OC.3 Data-Based Microfoundations

Microfoundation 1: link-specific measurement identifies marginals but not correlations. In many network applications, different links (i, j) are measured using distinct data sources, instruments, or experimental designs, so the DM can estimate each $\mathbf{m}_{ij}$ and $\mathbf{v}_{ij}^2$ from link-by-link variation but does not observe joint realizations that would identify covariances across (i, j) and (i, k) for fixed i and distinct j, k. The resulting statistical information disciplines the diagonal of $\mathbf{B}_i$ but leaves the off-diagonals largely unrestricted, motivating the use of $\mathcal{B}_i$.

Microfoundation 2: limited overlap in panels or rotating samples. Even when all links are conceptually observed in a single dataset, limitations such as short panels, rotating samples, or missing-by-design observations may imply that, for each (i, j), there is enough repeated variation to estimate $\mathrm{Var}(\mathbf{G}_{ij})$ but insufficient overlap to estimate $\mathrm{Cov}(\mathbf{G}_{ij}, \mathbf{G}_{ik})$ with precision. In such settings, treating off-diagonal dependence as ambiguous while trusting the marginal second moments is a disciplined reduced-form approximation, and worst-case evaluation over $\mathcal{B}_i$ corresponds to a conservative policy criterion that is valid across dependence patterns consistent with the data.

Microfoundation 3: latent common shocks generate comovement that is hard to discipline. Flexible structural sources of dependence are latent common shocks affecting multiple links toward the same receiver i. For instance, represent row-i deviations as $(\mathbf{U}_i)_j = a_{ij} f_i + u_{ij}$ for $j \in N$, where f_i is a receiver-i common shock with $\mathrm{Var}(f_i) = \sigma_i^2$, a_{ij} are link-specific loadings, and u_{ij} is idiosyncratic noise with $\mathrm{E}[u_{ij}] = 0$ and $\mathrm{Var}(u_{ij}) = \mathbf{v}_{ij}^2 - a_{ij}^2 \sigma_i^2$, with $a_{ij}^2 \sigma_i^2 \leq \mathbf{v}_{ij}^2$. Many combinations of

[20] For related approaches to robust decision making with partially identified models, see, for example, Stoye (2012); Montiel Olea et al. (2025).

$(a_{ij})_{j \in N}$ and σ_i^2 are consistent with the same marginal variances $\mathbf{v}_{ij}^2$ but imply different covariance patterns $\mathrm{Cov}(\mathbf{G}_{ij}, \mathbf{G}_{ik}) = a_{ij} a_{ik} \sigma_i^2$ across (j, k). When the DM can estimate $\mathbf{v}_{ij}^2$ but cannot credibly bound the strength of the common component, $\mathcal{B}_i$ captures the implied ambiguity without committing to a particular factor specification.

OC.4 Why Receiver-Specific Ambiguity and Aggregation are Natural

The object $\mathbf{U}_i$ aggregates the uncertainty relevant for receiver i because it determines the random component of $(\mathbf{Gx})_i$. Accordingly, the maintained information and the ambiguity are naturally formulated row-by-row: the DM's knowledge concerns marginal moments of each $\mathbf{G}_{ij}$, while the economically relevant unknowns are the covariances among links entering a common receiver. The aggregate set $\mathcal{B}$ arises from combining receiver-specific covariance choices $\mathbf{B}_i \in \mathcal{B}_i$ via $\mathbf{B} = \sum_{i=1}^n \mathbf{B}_i$, yielding a transparent interpretation of Nature's move: Nature selects a feasible dependence structure within each row, subject to the DM's trusted marginal second moments and covariance feasibility, to maximize the DM's expected loss under the chosen intervention.